

LLM-Based Triage of GPU Numerical Failures from Public Issue Reports

Paper #XXX
Anonymous Submission
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1 Motivation

GPU software stacks now combine tensor libraries, generated kernels, compiler passes, mixed-precision execution, and hardware-specific runtime behavior. When these systems fail, the first useful evidence often appears in public issue trackers as noisy natural-language reports: NaNs, Inf values, overflow, wrong answers, dtype conversion failures, compiler crashes, and performance regressions are frequently mixed in the same thread. This creates a practical systems problem. Maintainers need to route reports to the right experts, and researchers need reusable traces of reliability failures, but weak labels from keyword search are not trustworthy enough to support either goal.

2 Contribution

We present GPURTriage, a human-labeled benchmark and LLM-based triage system for GPU numerical-failure reports mined from public repositories. The benchmark contains 1,191 human-labeled issues from eight GPU-related projects spanning compiler, runtime, tensor, and data-processing software. Each issue receives one primary label from a seven-class taxonomy covering invalid values, overflow/underflow, precision or tolerance failures, dtype or casting failures, crash/compile failures, performance-only reports, and non-numerical reports. The artifact also includes annotation instructions, blind-audit files, adjudication logs, taxonomy repair logs, model outputs, evaluation tables, an error appendix, and root-cause evidence files.

3 Method and Results

GPURTriage separates data mining from ground truth. Candidate issues are first retrieved with terms such as nan, inf, overflow, dtype, precision, and wrong result. These candidate labels are treated only as weak supervision. Human annotation then repairs the label space and creates the canonical benchmark. A blind audit over 119 rows achieves 80.7% agreement and Cohen’s $\kappa = 0.721$; subsequent adjudication passes apply 419 human updates and change 211 primary labels.

This design targets an operational systems workflow rather than a generic text classification benchmark. A maintainer-facing assistant should identify which reports are likely numerical failures, route them into the right queue, and abstain when the evidence is ambiguous. For that reason, we evaluate both full-coverage prediction and

Table 1. Key ATC submission results.

Setting	Accuracy	Macro F1
Weak retrieval labels	0.505	0.426
Voting ensemble	0.738	0.712
FLAN-T5-base LLM	0.805	0.778
LLM/ensemble agreement	0.893	0.844
Leave-one-repository-out	0.668	0.648

selective prediction. The latter is important because wrong labels are costly: a dtype report routed as a compiler crash or a non-numerical API question routed as a correctness bug wastes maintainer time.

The system evaluates both deterministic models and a fine-tuned LLM. Weak retrieval labels reach only 50.5% accuracy, demonstrating that raw issue-search labels are not reliable ground truth. A deterministic voting ensemble reaches 73.8% accuracy and 0.712 macro F1. A fine-tuned FLAN-T5-base classifier reaches 80.5% held-out accuracy and 0.778 macro F1. For high-confidence triage, an LLM/deterministic agreement mode reaches 89.3% accuracy and 0.844 macro F1 at 68.3% coverage. Chronological evaluation reaches 76.6% accuracy, while leave-one-repository-out evaluation reaches 66.8%, exposing cross-project transfer as the main remaining challenge.

4 Why This Fits ATC

The work is a systems paper because it studies practical failure reports from real GPU software ecosystems and produces an artifact for bug finding, analyzing, and troubleshooting. Its main result is not only that an LLM can classify issue text; it is that reliable triage requires curated systems data, auditable label repair, cross-repository evaluation, and selective confidence mechanisms. The benchmark makes previously unavailable GPU numerical-failure traces available for future systems reliability work. The full paper discusses the system design, data construction, model evaluation, cross-repository weaknesses, and limitations needed to judge whether the results support the claim.

The expected ATC contribution is therefore an artifact-backed systems result: public GPU issue reports can be turned into a reusable triage benchmark, but only after weak labels are repaired and cross-repository transfer is measured

explicitly. The paper does not claim definitive root-cause diagnosis from issue text alone. Instead, it establishes report-level triage as a measurable systems task and provides the data needed for future root-cause and debugging research.

References

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